

Table 1: Aggregate win counts comparing the inverse-rate regularizer to the baseline (SGD + L2 + augmentation) across 19 architectures on CIFAR-10 (top) and CIFAR-100 (bottom). Each row corresponds to a specific setting of the hyper-parameters ρ and λ^* . “Best” denotes per-model selection based on validation NLL. For each metric, the table reports how many times the regularized model outperformed the baseline across the 19 model architectures.

Architecture	ρ	λ^*	Top1-Acc	Top5-Acc	NLL	Variance	ECE
CIFAR10(LAMB.WD)	Best	Best	4/19	9/19	18/19	19/19	17/19
	0.0	Best	0/19	5/19	9/19	19/19	19/19
	0.5	Best	2/19	12/19	17/19	19/19	18/19
	1.0	Best	6/19	10/19	17/19	19/19	16/19
CIFAR100(LAMB.WD)	Best	Best	3/19	10/19	18/19	19/19	18/19
	0.0	Best	0/19	8/19	6/19	17/19	16/19
	0.5	Best	2/19	13/19	18/19	17/19	14/19
	1.0	Best	4/19	8/19	18/19	19/19	18/19
CIFAR-10(celoss)	Best	Best	7/19	11/19	18/19	19/19	17/19
	0.0	Best	0/19	4/19	10/19	19/19	19/19
	0.5	Best	0/19	9/19	17/19	19/19	19/19
	1.0	Best	13/19	11/19	16/19	18/19	17/19
CIFAR100(celoss)	Best	Best	3/19	14/19	19/19	16/19	18/19
	0.0	Best	0/19	5/19	8/19	18/19	18/19
	0.5	Best	3/19	12/19	18/19	15/19	13/19
	1.0	Best	8/19	18/19	19/19	16/19	18/19

Table 2: Comparison .

CIFAR-10lamb wd	Top1-Acc ($\uparrow$)		Top5-Acc ($\uparrow$)		NLL ($\downarrow$)		Variance ($\downarrow$)		ECE ($\downarrow$)	
	Base	Ours	Base	Ours	Base	Ours	Base	Ours	Base	Ours
mobilenetv2_x0.5	0.721	0.694	0.922	0.922	1.176	1.073	4.705	2.609	0.119	0.030
mobilenetv2_x0.75	0.739	0.713	0.927	0.926	1.097	1.034	4.395	2.854	0.110	0.048
mobilenetv2_x1.0	0.744	0.718	0.932	0.929	1.063	1.045	4.414	3.128	0.111	0.060
mobilenetv2_x1.4	0.767	0.735	0.936	0.931	0.981	0.999	3.826	3.071	0.091	0.061
repvgg_a0	0.745	0.729	0.925	0.930	1.066	1.011	4.004	2.366	0.089	0.029
repvgg_a1	0.762	0.775	0.933	0.940	0.988	0.913	3.349	2.777	0.074	0.060
repvgg_a2	0.766	0.775	0.938	0.949	0.965	0.846	2.929	2.676	0.059	0.058
resnet20	0.685	0.677	0.908	0.916	1.266	1.128	4.536	2.526	0.104	0.015
resnet32	0.691	0.699	0.906	0.918	1.350	1.098	6.024	3.030	0.145	0.049
resnet44	0.703	0.698	0.912	0.919	1.374	1.107	6.578	3.383	0.150	0.075
resnet56	0.718	0.711	0.913	0.923	1.364	1.105	6.616	3.657	0.148	0.088
shufflenetv2_x0.5	0.666	0.664	0.901	0.906	1.333	1.215	4.982	3.106	0.129	0.054
shufflenetv2_x1.0	0.719	0.712	0.915	0.924	1.189	1.109	4.533	3.754	0.111	0.093
shufflenetv2_x1.5	0.746	0.739	0.920	0.927	1.076	1.024	3.799	3.292	0.080	0.079
shufflenetv2_x2.0	0.749	0.747	0.932	0.928	1.014	0.986	3.371	3.114	0.069	0.071
vgg11_bn	0.694	0.656	0.882	0.874	1.546	1.388	7.663	4.303	0.156	0.082
vgg13_bn	0.737	0.701	0.909	0.900	1.317	1.207	6.650	4.016	0.130	0.084
vgg16_bn	0.730	0.701	0.904	0.883	1.665	1.390	10.546	6.084	0.193	0.128
vgg19_bn	0.740	0.680	0.901	0.867	1.809	1.484	13.251	6.594	0.201	0.115
All Models	15/19	4/19	10/19	9/19	1/19	18/19	0/19	19/19	2/19	17/19
CIFAR-100lamb wd	Top1-Acc ($\uparrow$)		Top5-Acc ($\uparrow$)		NLL ($\downarrow$)		Variance ($\downarrow$)		ECE ($\downarrow$)	
	Base	Ours	Base	Ours	Base	Ours	Base	Ours	Base	Ours
mobilenetv2_x0.5	0.926	0.923	0.997	0.997	0.281	0.247	1.318	0.861	0.044	0.030
mobilenetv2_x0.75	0.941	0.926	0.998	0.999	0.248	0.230	1.297	0.721	0.036	0.027
mobilenetv2_x1.0	0.939	0.932	0.998	0.997	0.250	0.232	1.281	0.819	0.040	0.027
mobilenetv2_x1.4	0.948	0.943	0.998	0.998	0.217	0.224	1.117	1.100	0.031	0.032
repvgg_a0	0.947	0.947	0.998	0.998	0.260	0.229	1.575	1.212	0.037	0.034
repvgg_a1	0.946	0.942	0.998	0.996	0.248	0.221	1.426	0.995	0.036	0.034
repvgg_a2	0.952	0.951	0.998	0.998	0.214	0.192	1.155	0.937	0.032	0.026
resnet20	0.923	0.928	0.998	0.998	0.286	0.236	1.276	0.774	0.043	0.024
resnet32	0.940	0.937	0.997	0.998	0.275	0.234	1.632	1.028	0.040	0.035
resnet44	0.942	0.941	0.998	0.997	0.253	0.232	1.394	1.088	0.037	0.035
resnet56	0.938	0.942	0.997	0.998	0.294	0.235	1.792	1.186	0.045	0.037
shufflenetv2_x0.5	0.901	0.914	0.995	0.997	0.363	0.297	1.615	1.139	0.058	0.034
shufflenetv2_x1.0	0.933	0.924	0.997	0.997	0.279	0.265	1.489	0.984	0.039	0.036
shufflenetv2_x1.5	0.942	0.937	0.997	0.998	0.245	0.241	1.275	1.160	0.034	0.035
shufflenetv2_x2.0	0.938	0.938	0.997	0.997	0.250	0.241	1.221	1.183	0.038	0.034
vgg11_bn	0.926	0.900	0.997	0.994	0.344	0.309	1.906	0.822	0.053	0.027
vgg13_bn	0.941	0.945	0.997	0.997	0.277	0.239	1.579	1.219	0.042	0.036
vgg16_bn	0.941	0.937	0.997	0.997	0.312	0.274	1.987	1.370	0.046	0.045
vgg19_bn	0.937	0.934	0.996	0.996	0.339	0.302	2.143	1.577	0.051	0.047
All Models	16/19	3/19	9/19	10/19	1/19	18/19	0/19	19/19	1/19	18/19

Table 3: Comparison of the baseline (**Base**) method (SGD + L2 + data augmentation) versus our inverse-rate regularizer

CIFAR-10 ce loss	Top1-Acc ($\uparrow$)		Top5-Acc ($\uparrow$)		NLL ($\downarrow$)		Variance ($\downarrow$)		ECE ($\downarrow$)	
	Base	Ours	Base	Ours	Base	Ours	Base	Ours	Base	Ours
mobilenetv2_x0.5	0.926	0.923	0.997	0.997	0.281	0.247	1.318	0.861	0.044	0.030
mobilenetv2_x0.75	0.941	0.944	0.998	0.997	0.248	0.235	1.297	1.176	0.036	0.033
mobilenetv2_x1.0	0.939	0.944	0.998	0.998	0.250	0.220	1.281	1.092	0.040	0.033
mobilenetv2_x1.4	0.948	0.946	0.998	0.998	0.217	0.223	1.117	1.080	0.031	0.033
repvgg_a0	0.947	0.939	0.998	0.997	0.260	0.240	1.575	1.173	0.037	0.034
repvgg_a1	0.946	0.945	0.998	0.998	0.248	0.216	1.426	1.051	0.036	0.030
repvgg_a2	0.952	0.949	0.998	0.998	0.214	0.207	1.155	1.054	0.032	0.029
resnet20	0.923	0.920	0.998	0.998	0.286	0.247	1.276	0.861	0.043	0.032
resnet32	0.940	0.933	0.997	0.998	0.275	0.233	1.632	0.902	0.040	0.028
resnet44	0.942	0.940	0.998	0.998	0.253	0.226	1.394	1.017	0.037	0.030
resnet56	0.938	0.938	0.997	0.998	0.294	0.240	1.792	1.099	0.045	0.033
shufflenetv2_x0.5	0.901	0.914	0.995	0.997	0.363	0.297	1.615	1.139	0.058	0.034
shufflenetv2_x1.0	0.933	0.934	0.997	0.998	0.279	0.269	1.489	1.379	0.039	0.039
shufflenetv2_x1.5	0.942	0.942	0.997	0.998	0.245	0.234	1.275	1.157	0.034	0.034
shufflenetv2_x2.0	0.938	0.944	0.997	0.998	0.250	0.237	1.221	1.159	0.038	0.036
vgg11_bn	0.926	0.919	0.997	0.997	0.344	0.311	1.906	1.394	0.053	0.050
vgg13_bn	0.941	0.947	0.997	0.998	0.277	0.245	1.579	1.379	0.042	0.038
vgg16_bn	0.941	0.934	0.997	0.998	0.312	0.281	1.987	1.370	0.046	0.044
vgg19_bn	0.937	0.931	0.996	0.997	0.339	0.284	2.143	1.393	0.051	0.044

Table 4: Comparison

CIFAR-100 ce loss	Top1-Acc ($\uparrow$)		Top5-Acc ($\uparrow$)		NLL ($\downarrow$)		Variance ($\downarrow$)		ECE ($\downarrow$)	
	Base	Ours	Base	Ours	Base	Ours	Base	Ours	Base	Ours
mobilenetv2_x0.5	0.721	0.699	0.922	0.922	1.176	1.064	4.705	2.826	0.119	0.047
mobilenetv2_x0.75	0.739	0.704	0.927	0.927	1.097	1.049	4.395	2.742	0.110	0.042
mobilenetv2_x1.0	0.744	0.725	0.932	0.930	1.063	1.021	4.414	3.255	0.111	0.069
mobilenetv2_x1.4	0.767	0.766	0.936	0.937	0.981	0.969	3.826	3.864	0.091	0.086
repvgg_a0	0.745	0.744	0.925	0.927	1.066	1.031	4.004	3.609	0.089	0.085
repvgg_a1	0.762	0.754	0.933	0.933	0.988	0.980	3.349	3.374	0.074	0.073
repvgg_a2	0.766	0.773	0.938	0.942	0.965	0.912	2.929	2.762	0.059	0.059
resnet20	0.685	0.675	0.908	0.916	1.266	1.133	4.536	2.942	0.104	0.059
resnet32	0.691	0.699	0.906	0.919	1.350	1.127	6.024	3.654	0.145	0.077
resnet44	0.703	0.693	0.912	0.920	1.374	1.106	6.578	3.031	0.150	0.054
resnet56	0.718	0.690	0.913	0.912	1.364	1.123	6.616	3.137	0.148	0.054
shufflenetv2_x0.5	0.666	0.667	0.901	0.904	1.333	1.208	4.982	3.059	0.129	0.048
shufflenetv2_x1.0	0.719	0.701	0.915	0.919	1.189	1.107	4.533	3.449	0.111	0.085
shufflenetv2_x1.5	0.746	0.739	0.920	0.927	1.076	1.024	3.799	3.292	0.080	0.079
shufflenetv2_x2.0	0.749	0.747	0.932	0.936	1.014	0.983	3.371	3.507	0.069	0.087
vgg11_bn	0.694	0.686	0.882	0.898	1.546	1.420	7.663	6.226	0.156	0.146
vgg13_bn	0.737	0.736	0.909	0.909	1.317	1.277	6.650	6.251	0.130	0.130
vgg16_bn	0.730	0.706	0.904	0.893	1.665	1.361	10.546	6.349	0.193	0.144
vgg19_bn	0.740	0.682	0.901	0.882	1.809	1.393	13.251	5.798	0.201	0.123

Table 5: Aggregate win counts

Architecture	ρ	λ^*	Top1-Acc	Top5-Acc	NLL	Variance	ECE
CIFAR10_lamb_wd	Best	Best	4/19	9/19	18/19	19/19	17/19
	0.0	Best	0/19	5/19	9/19	19/19	19/19
	0.5	Best	2/19	12/19	17/19	19/19	18/19
	1.0	Best	6/19	10/19	17/19	19/19	16/19
	Best	0.001	0/19	0/19	0/19	19/19	19/19
	Best	0.01	0/19	0/19	0/19	19/19	19/19
	Best	0.1	0/19	0/19	7/19	19/19	19/19
	Best	0.5	3/19	10/19	17/19	19/19	19/19
	Best	1.0	3/19	8/19	18/19	19/19	17/19
CIFAR100_lamb_wd	Best	Best	3/19	10/19	18/19	19/19	18/19
	0.0	Best	0/19	8/19	6/19	17/19	16/19
	0.5	Best	2/19	13/19	18/19	17/19	14/19
	1.0	Best	4/19	8/19	18/19	19/19	18/19
	Best	0.001	0/19	0/19	0/19	19/19	19/19
	Best	0.01	0/19	0/19	0/19	19/19	19/19
	Best	0.1	0/19	1/19	11/19	19/19	16/19
	Best	0.5	5/19	13/19	18/19	19/19	16/19
	Best	1.0	3/19	12/19	15/19	13/19	14/19
cifar10_celoss	Best	Best	7/19	11/19	18/19	19/19	17/19
	0.0	Best	0/19	4/19	10/19	19/19	19/19
	0.5	Best	0/19	9/19	17/19	19/19	19/19
	1.0	Best	13/19	11/19	16/19	18/19	17/19
	Best	0.001	5/19	9/19	14/19	16/19	15/19
	Best	0.01	7/19	11/19	16/19	17/19	16/19
	Best	0.1	4/19	13/19	18/19	19/19	16/19
	Best	0.5	3/19	7/19	16/19	18/19	14/19
	Best	1.0	3/19	8/19	18/19	19/19	17/19
CIFAR100_celoss	Best	Best	3/19	14/19	19/19	16/19	18/19
	0.0	Best	0/19	5/19	8/19	18/19	18/19
	0.5	Best	3/19	12/19	18/19	15/19	13/19
	1.0	Best	8/19	18/19	19/19	16/19	18/19
	Best	0.001	4/19	11/19	18/19	16/19	17/19
	Best	0.01	3/19	11/19	16/19	15/19	16/19
	Best	0.1	4/19	11/19	17/19	17/19	15/19
	Best	0.5	3/19	10/19	15/19	14/19	14/19
	Best	1.0	3/19	12/19	15/19	13/19	14/19